

$$1. X \text{ matrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Diya Jain

U81673988

$$\det \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 - 1$$

$$\lambda^2 - 1 = 0$$

$$(\lambda - 1)(\lambda + 1) = 0$$

$$\lambda = 1, \lambda = -1$$

• $\lambda = -1$

$$\begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow[R_2 \rightarrow R_2 - R_1]{} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{ if we take } x_1 = -t \text{ and } x_2 = t$$

$$\bar{x} = \begin{bmatrix} -t \\ t \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \leftarrow \text{eigenvector}$$

• $\lambda = 1$

$$\begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow[R_2 \rightarrow R_2 - R_1]{R_1 \rightarrow -R_1} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, x_1 = t, x_2 = t$$

$$\bar{x} = \begin{bmatrix} t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \leftarrow \text{eigenvector}$$

$$Z\text{-matrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\det \begin{bmatrix} 1-\lambda & 0 \\ 0 & -\lambda-1 \end{bmatrix} = (1-\lambda)(-\lambda-1)$$

$$(1-\lambda)(-\lambda-1) = 0$$

$$\lambda = 1 \quad \lambda = -1$$

$$\bullet \lambda = 1$$

$$\begin{bmatrix} 1-\lambda & 0 \\ 0 & -\lambda-1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \xrightarrow[R_1 \rightarrow -\frac{R_1}{2}]{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_1 = t, \quad x_2 = 0$$

$$\bar{x} = \begin{bmatrix} t \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} t = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \leftarrow \text{eigenvector}$$

$$\bullet \lambda = -1$$

$$\begin{bmatrix} 1-\lambda & 0 \\ 0 & -\lambda-1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1/2} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_2 = t, \quad x_1 = 0$$

$$\bar{x} = \begin{bmatrix} 0 \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} t = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \leftarrow \text{eigenvector}$$

X gate: It flips state of qubit from $|0\rangle$ to $|1\rangle$ and from $|1\rangle$ to $|0\rangle$.

Z gate: It phases the state of the qubit, leaving $|0\rangle$ as it is and multiplying -1 with the $|1\rangle$.

$$2.a. \quad |\gamma_0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \quad |\gamma_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$H|\gamma_0\rangle = H \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) = \frac{1}{2} (|10\rangle + |11\rangle + |0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} |0\rangle$$

$$H|\gamma_1\rangle = H \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) = \frac{1}{2} (|10\rangle + |11\rangle + |0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} |0\rangle$$

$$HH|\gamma_0\rangle = \frac{1}{\sqrt{2}} H|0\rangle = \frac{1}{2} (|0\rangle + |1\rangle)$$

$$HH|\gamma_1\rangle = \frac{1}{\sqrt{2}} H|0\rangle = \frac{1}{2} (|0\rangle + |1\rangle)$$

The probability is $\frac{1}{2}$ for measuring $|0\rangle$ and $|1\rangle$ each

$$b. |\psi_0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \quad \text{and} \quad |\psi_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$\frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

$$H_0 H_1 \left(\frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right) = |0\rangle \otimes |1\rangle$$

$$CNOT (|0\rangle \otimes |1\rangle) = |0\rangle \otimes |1\rangle$$

$$H_0 H_1 (|0\rangle \otimes |1\rangle) = \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

$$= \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

The probability is $\frac{1}{2}$ for measuring $|0\rangle$ and $|1\rangle$ each

$$c. |\psi_0\rangle = |0\rangle \quad \text{and} \quad |\psi_1\rangle = |1\rangle$$

$$H_0 H_1 (|01\rangle) = \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

$$CNOT \left(\frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right) = \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |11\rangle - \frac{1}{2} |10\rangle$$

$$H_0 H_1 \left(\frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |11\rangle - \frac{1}{2} |10\rangle \right)$$

$$= H O H I \left(\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right) = |11\rangle$$

$|11\rangle$ will be the 100% probability.

d. $|\psi_0\rangle = |1\rangle$ and $|\psi_1\rangle = |1\rangle$

$$H O H I (|111\rangle) = \frac{1}{2} |100\rangle - \frac{1}{2} |101\rangle + \frac{1}{2} |111\rangle - \frac{1}{2} |110\rangle$$

$$CNOT \left(\frac{1}{2} |100\rangle - \frac{1}{2} |101\rangle + \frac{1}{2} |111\rangle - \frac{1}{2} |110\rangle \right) = \frac{1}{2} |100\rangle - \frac{1}{2} |101\rangle + \frac{1}{2} |110\rangle - \frac{1}{2} |111\rangle$$

$$H O H I \left(\frac{1}{2} |100\rangle - \frac{1}{2} |101\rangle + \frac{1}{2} |110\rangle - \frac{1}{2} |111\rangle \right) = H O H I \left(\frac{1}{\sqrt{2}} (|10\rangle + |11\rangle) \right) \otimes \frac{1}{\sqrt{2}} (|10\rangle - |11\rangle)$$

$$= |01\rangle$$

3.

When terms cancel each other out in quantum mechanics, it's often referred to as interference or destructive interference. Interference occurs when different pathways or amplitudes in a quantum system combine in such a way that they partially or completely cancel each other out at certain points or states, leading to observable effects like reduced probabilities or zero probability of certain outcomes.

4.

Qiskit

Read only

Open in Quantum Lab

```
1 from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
2 from numpy import pi
3
4 qreg_q = QuantumRegister(2, 'q')
5 creg_c = ClassicalRegister(2, 'c')
6 circuit = QuantumCircuit(qreg_q, creg_c)
7
8 circuit.reset(qreg_q[0])
9 circuit.reset(qreg_q[1])
10 circuit.x(qreg_q[1])
11 circuit.barrier(qreg_q[0])
12 circuit.barrier(qreg_q[1])
13 circuit.h(qreg_q[0])
14 circuit.h(qreg_q[1])
15 circuit.barrier(qreg_q[0])
16 circuit.barrier(qreg_q[1])
17 circuit.cx(qreg_q[0], qreg_q[1])
18 circuit.x(qreg_q[1])
19 circuit.barrier(qreg_q[0])
20 circuit.barrier(qreg_q[1])
21 circuit.h(qreg_q[0])
22 circuit.measure(qreg_q[0], creg_c[0])
23 # @columns [0,0,1,2,2,3,3,4,4,5,6,7,7,8,9]
```

Operations

Search

H

$\oplus$

$\otimes$

$\otimes$

$\otimes$

I

T

S

Z

$T^\dagger$

$S^\dagger$

P

RZ

$\curvearrowright^2$

$|0\rangle$

$\vdots$

$\bullet$

if

$\sqrt{X}$

$\sqrt{X}^\dagger$

q[0]

$|0\rangle$

H

H

$\curvearrowright^2$

q[1]

$|0\rangle$

$\oplus$

H

$\oplus$

$\oplus$

c2

0

Probabilities

Probability (% of 1024 shots)

100

80

60

40

20

0

00

01

10

11

Computational basis states

Q-sphere

$\pi/2$

Phase

0

$3\pi/2$

101

111

☒ State ☐ Phase angle

Terms

Privacy

Cookie preferences

Support

Qiskit

Read only

Open in Quantum Lab

```
1 from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
2 from numpy import pi
3
4 qreg_q = QuantumRegister(2, 'q')
5 creg_c = ClassicalRegister(2, 'c')
6 circuit = QuantumCircuit(qreg_q, creg_c)
7
8 circuit.reset(qreg_q[0])
9 circuit.reset(qreg_q[1])
10 circuit.x(qreg_q[1])
11 circuit.barrier(qreg_q[0])
12 circuit.barrier(qreg_q[1])
13 circuit.h(qreg_q[0])
14 circuit.h(qreg_q[1])
15 circuit.barrier(qreg_q[0])
16 circuit.barrier(qreg_q[1])
17 circuit.cx(qreg_q[0], qreg_q[1])
18 circuit.x(qreg_q[1])
19 circuit.cx(qreg_q[0], qreg_q[1])
20 circuit.barrier(qreg_q[0])
21 circuit.barrier(qreg_q[1])
22 circuit.h(qreg_q[0])
23 circuit.measure(qreg_q[0], creg_c[0])
24 # @columns [0,0,1,2,2,3,3,4,4,5,6,7,8,8,9,10]
```

Operations

Search

H

$\oplus$

$\otimes$

$\otimes$

$\otimes$

I

T

S

Z

$T^\dagger$

$S^\dagger$

P

RZ

$\curvearrowright^2$

$|0\rangle$

$\vdots$

$\bullet$

if

$\sqrt{X}$

$\sqrt{X}^\dagger$

Y

RX

RY

RXX

RZZ

q[0]

$|0\rangle$

H

H

$\curvearrowright^2$

q[1]

$|0\rangle$

$\oplus$

H

$\oplus$

$\oplus$

c2

0

Probabilities

Probability (% of 1024 shots)

100

80

60

40

20

0

00

01

10

11

Computational basis states

Q-sphere

$\pi/2$

Phase

0

$3\pi/2$

100

110

☒ State ☐ Phase angle

Qiskit

Read only

Open in Quantum Lab

```
1 from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
2 from numpy import pi
3
4 qreg_q = QuantumRegister(2, 'q')
5 creg_c = ClassicalRegister(2, 'c')
6 circuit = QuantumCircuit(qreg_q, creg_c)
7
8 circuit.reset(qreg_q[0])
9 circuit.reset(qreg_q[1])
10 circuit.x(qreg_q[1])
11 circuit.barrier(qreg_q[0])
12 circuit.barrier(qreg_q[1])
13 circuit.h(qreg_q[0])
14 circuit.h(qreg_q[1])
15 circuit.barrier(qreg_q[0])
16 circuit.barrier(qreg_q[1])
17 circuit.cx(qreg_q[0], qreg_q[1])
18 circuit.barrier(qreg_q[0])
19 circuit.barrier(qreg_q[1])
20 circuit.h(qreg_q[0])
21 circuit.measure(qreg_q[0], creg_c[0])
22 # @columns [0,0,1,2,2,3,3,4,4,5,6,6,7,8]
```

Operations

Search

H

$\oplus$

$\otimes$

$\otimes$

$\otimes$

I

T

S

Z

$T^\dagger$

$S^\dagger$

P

RZ

$\curvearrowright^2$

$|0\rangle$

$\vdots$

$\bullet$

if

$\sqrt{X}$

$\sqrt{X}^\dagger$

Y

RX

RY

RXX

RZZ

q[0]

$|0\rangle$

H

H

$\curvearrowright^2$

q[1]

$|0\rangle$

$\oplus$

H

$\oplus$

c2

0

Probabilities

Probability (% of 1024 shots)

100

80

60

40

20

0

00

01

10

11

Computational basis states

Q-sphere

$\pi/2$

Phase

0

$3\pi/2$

101

111

☒ State ☐ Phase angle